

# Deployment Guide

Smart Analysis Reporter

Prepared by Kavın Ganapathy

Student ID: 100008820

Date: 2026-06-07

# Table of Contents

|                                     |   |
|-------------------------------------|---|
| 1. Overview                         | 3 |
| 2. Flow A — Push to GitHub          | 4 |
| 3. Flow B — Deploy to a Public Link | 5 |
| 4. Updating the Live App            | 6 |
| 5. Troubleshooting                  | 7 |
| 6. Quick Temporary Link             | 8 |

# 1. Overview

This guide explains how to publish **Smart Analysis Reporter** as a live, public web application. It is organised as two independent process flows:

- **Flow A — Push the project to GitHub.**
- **Flow B — Deploy from GitHub to a public link** (Streamlit Community Cloud).

Target repository: <https://github.com/TejalMenezes/analysis-toolkit-v1> (branch **main**).

## 1.1 Pre-flight checklist

These are already prepared in the project:

- requirements.txt lists every dependency.
- .gitignore excludes the virtual environment, caches, logs and local settings.
- The default dataset is committed at data/student\_dataset.csv, so the deployed app needs no Kaggle login.
- The application entry point is app.py.
- The Git repository is initialised and committed locally.

## 2. Flow A — Push to GitHub

Run these commands from the project root. They point the local repository at GitHub and upload the code.

```
# 1. point the local repo at your GitHub repo (once)
git remote add origin https://github.com/TejalMenezes/analysis-toolkit-v1.git

# 2. make sure you are on main
git branch -M main

# 3. stage and commit any pending changes
git add .
git commit -m "Smart Analysis Reporter"

# 4. push
git push -u origin main
```

### 2.1 If the remote already exists

If Git reports “remote origin already exists”, update its URL instead:

```
git remote set-url origin https://github.com/TejalMenezes/analysis-toolkit-v1.git
git push -u origin main
```

### 2.2 Authentication

GitHub no longer accepts your account password on the command line — use a **Personal Access Token (PAT)**:

- In GitHub, go to Settings → Developer settings → Personal access tokens → Tokens (classic).
- Generate a new token with the **repo** scope and copy it.
- When *git push* asks for a password, paste the token.
- Alternatively, use GitHub Desktop or the GitHub CLI (*gh auth login*), which handle authentication for you.

**Note on multiple accounts.** If the machine has cached credentials for a different GitHub account, the push will be rejected with a 403 error. Clear the stored credential (Windows Credential Manager → remove the github.com entry) or push using a token for the correct account.

### 2.3 Verify

Refresh <https://github.com/TejalMenezes/analysis-toolkit-v1> in your browser — all the project files should now appear.

## 3. Flow B — Deploy to a Public Link

Once the repository is on GitHub, Streamlit Community Cloud builds and hosts the app for free — there is no server to manage.

### 3.1 Steps

- Go to <https://share.streamlit.io> and sign in with GitHub (authorise repo access).
- Click “Create app” → “Deploy a public app from GitHub.”
- Repository: **TejalMenezes/analysis-toolkit-v1**; Branch: **main**; Main file path: **app.py**.
- Click “Deploy.”
- Wait for the first build — it installs requirements.txt (typically 2–5 minutes; the log streams live).
- When the build finishes you receive your public URL, for example <https://analysis-toolkit-v1.streamlit.app> — that is the link to share.

### 3.2 After deployment

- Every push to main automatically redeploys the app — no extra steps.
- Manage, reboot or view logs from the dashboard at <https://share.streamlit.io>.
- If a build fails, open the log — it is almost always a missing line in requirements.txt.

## 4. Updating the Live App

To publish a change after the app is live, commit and push — Streamlit Cloud detects the push and rebuilds automatically.

```
git add .  
git commit -m "describe your change"  
git push
```

## 5. Troubleshooting

Common issues and how to resolve them:

| Symptom                          | Resolution                                                                      |
|----------------------------------|---------------------------------------------------------------------------------|
| remote origin already exists     | Run: <code>git remote set-url origin &lt;url&gt;</code> , then push.            |
| Push rejected (non-fast-forward) | Run: <code>git pull --rebase origin main</code> , then <code>git push</code> .  |
| Authentication fails on push     | Use a Personal Access Token as the password (see Section 2.2).                  |
| Cloud build fails on an import   | Add the missing package to <code>requirements.txt</code> , commit and push.     |
| App loads but reports no dataset | Confirm <code>data/student_dataset.csv</code> is committed (it is, by default). |

## 6. Quick Temporary Link

If you only need a public link for a short demo from your own machine (it lives only while the command runs), expose the local app with a tunnel:

```
# terminal 1
streamlit run app.py

# terminal 2
pip install pyngrok
python -c "from pyngrok import ngrok; print(ngrok.connect(8501))"
```

For a permanent link, use Flow B instead.